

PRANAVA PREETHIVARDHAN CHANDURI

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EDUCATION

Master of Science, Data Science — University of North Texas, Denton, TX

Expected May 2027

GPA: 4.0/4.0. Coursework: Deep Learning, Natural Language Processing, Machine Learning, Software Development for AI, Data Modeling

B.Tech, Computer Science & Engineering (AI & ML) — SRM University, India

May 2025

Coursework: Computer Vision, Image Processing, Operating Systems, Computer Architecture, Data Structures, Databases

TECHNICAL SKILLS

Languages: Python, C++, SQL, R

ML / Deep Learning: PyTorch, scikit-learn, XGBoost, LSTMs, CNNs, U-Net++, Sentence-Transformers; metrics (IoU, F1, Recall@K, MRR, nDCG)

AI / LLMs: LLMs, Transformers, Hugging Face, Fine-tuning (QLoRA), RAG, LangChain, FAISS, BM25, Cross-Encoder Reranking, Prompt Engineering

Data & Software Eng.: ETL pipelines, feature engineering, large-scale experimentation, REST APIs, FastAPI, Flask, Docker, CI/CD, Git, Linux, Unity 3D

Cloud & Databases: AWS EC2, MySQL, PostgreSQL, query optimization

RESEARCH & WORK EXPERIENCE

Graduate Research Assistant — Data Visualization & Extreme Reality Lab, University of North Texas

Aug 2025 – Mar 2026

- Contributed to an immersive VR/MR research system for cyber-situational awareness (PI: Dr. Sharad Sharma), turning abstract network-security data into navigable real-time 3D environments to improve analyst threat detection.
- Built Python/Pandas pipelines that cleaned and reshaped large-scale network-traffic datasets (CTU-13, CICIDS, NSL-KDD) into structured, time-sequenced JSON, extracting flows, IPs, timestamps, and labels as graph-ready inputs.
- Developed force-directed 3D node-link graphs and animated attack timelines in Unity 3D that render and update in real time, encoding command-and-control vs. normal traffic via node scale and color.
- Collaborated across an interdisciplinary UNT-GPREC research team through requirements gathering and iterative prototyping, surveying prior immersive-analytics literature to inform system design.

Software Developer Intern — MSRCosmos, Hyderabad, India

Jun 2023 – Aug 2023

- Built and deployed a full-stack e-commerce mobile application with RESTful APIs for high-volume transactional data; reduced database latency 25% via SQL query optimization and indexing.
- Integrated the app into CI/CD pipelines with automated testing and shipped on schedule within an Agile team through sprint planning, code reviews, and defect resolution.

OPEN SOURCE

DeepSpeed — Microsoft ([PR](#))

June 2026

- Contributed an open-source pull request to Microsoft's DeepSpeed — a widely adopted deep-learning optimization library for large-scale distributed and LLM training — adding a Trackio experiment-monitoring backend: implemented the TrackioMonitor class against the existing Monitor interface and integrated Hugging Face's WandB-compatible logging API, with rank-0-only initialization and graceful import handling consistent with the WandB/Comet/TensorBoard backends.

PROJECTS

MeetMind AI — Desktop Meeting Assistant ([GitHub](#))

Electron, React, FastAPI, Groq Llama 3.3

- Improved meeting-assistant signal-to-noise, surfacing nudges only when contextually warranted, by engineering a dual-speed "PRISM" intervention gate that weighs time remaining, unresolved items, topic drift, and deadlines to decide when to speak up versus stay silent, with per-interruption reasoning and tunable Quiet/Balanced/Active modes.
- Reduced manual post-meeting follow-up by auto-generating confidence-scored action items, decisions, summaries, and review-ready emails, building an LLM reasoning layer (Groq Llama 3.3) with FAISS episodic memory, a Neo4j habit graph, and an Accept/Edit/Dismiss loop that self-tunes thresholds, integrated with Google Calendar (OAuth) and SMTP over FastAPI REST/WebSocket endpoints.

Satellite Image Land Classification ([GitHub](#))

PyTorch, Flask

- Built a U-Net++ model (EfficientNet-B3 encoder) on the DeepGlobe dataset (803 annotated 50 cm/px images) for 5-class land-cover segmentation, achieving ~0.81 mean IoU and 90–96% per-class accuracy.
- Delivered an end-to-end Flask web app that segments uploaded satellite imagery and reports per-class area-distribution percentages; evaluated with IoU, precision, recall, F1, and confusion matrices.

Agentic RAG System ([GitHub](#))

FastAPI, FAISS, LangChain

- Built a production-grade agentic RAG system with a query planner and hybrid retrieval (BM25 + dense vectors via Reciprocal Rank Fusion, Cross-Encoder reranking), and a citation-validation layer that reduced unsupported responses and improved retrieval quality, evaluated using Recall@K, MRR, and nDCG; deployed via FastAPI with GitHub Actions CI.

Demand Forecasting ML Pipeline ([GitHub](#))

XGBoost, Time-Series

- Built an end-to-end forecasting pipeline on the Kaggle M5 dataset with engineered lag, rolling, calendar, price, and event features; trained an early-stopping XGBoost model under a leakage-free time-based split, beating a naïve baseline on MAE/RMSE to produce reproducible 28-day forecasts.

PUBLICATIONS & AWARDS

- "Comparative Analysis of Handwritten Digit Recognition Using MLP, CNN, and LeNet-5 Models." IEEE Conference Proceedings, 2025. [IEEE Xplore](#). **Best Paper Award**.
- "A Study on Blockchain Technology: Architecture, Consensus and Applications." ICICC 2025, Springer LNNS vol. 1432, pp. 349–360. [Springer](#).